

Election Analytics Platform for Portugal

Advanced Topics in Databases · Practical Assignment

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4

election cycles

2013 · 2017 · 2021 · 2025

308

municipalities mapped

PostgreSQL · PostGIS · Flask · D'Hondt

Goal & Scope



Database

Autárquicas
2013 · 2017
2021 · 2025



Spatial Layer

DGT CAOP
GeoPackages
loaded in PostGIS



ETL

Python loader
parameterized by
election + source

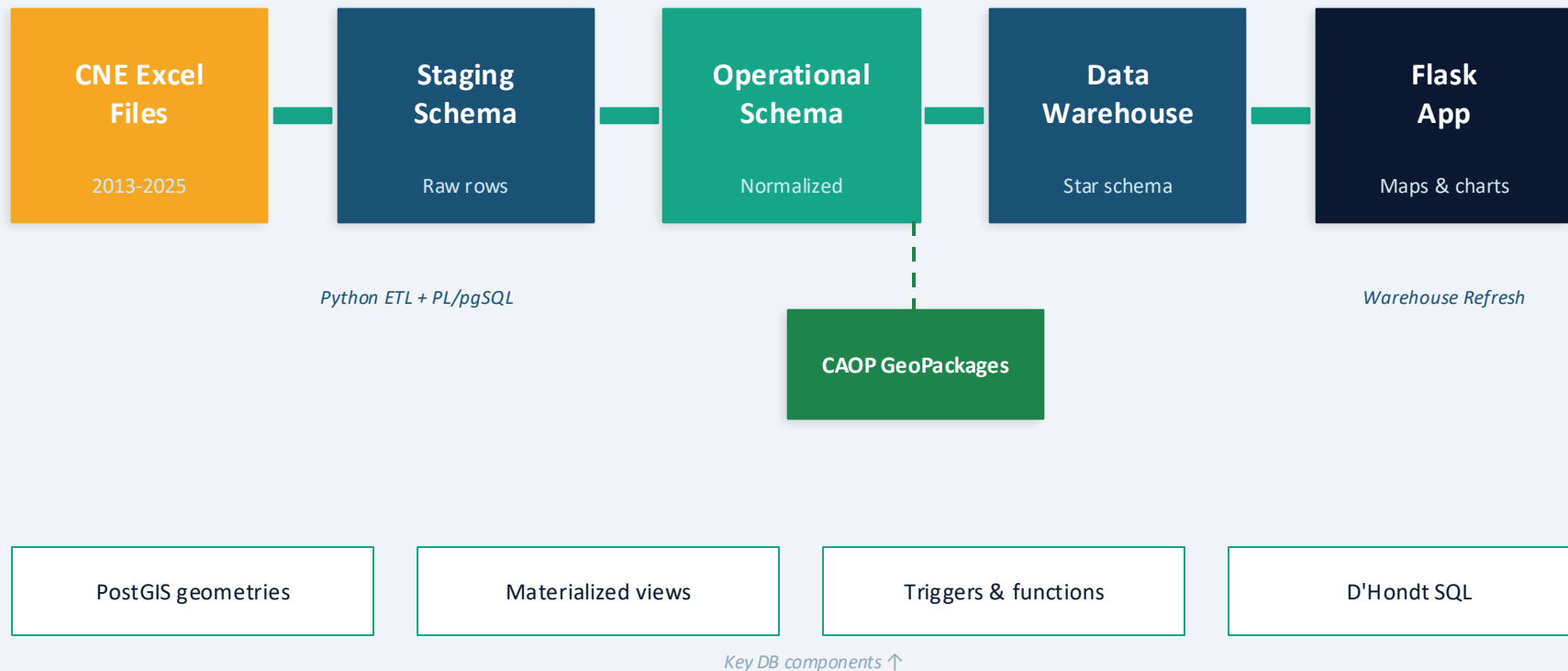


Flask App

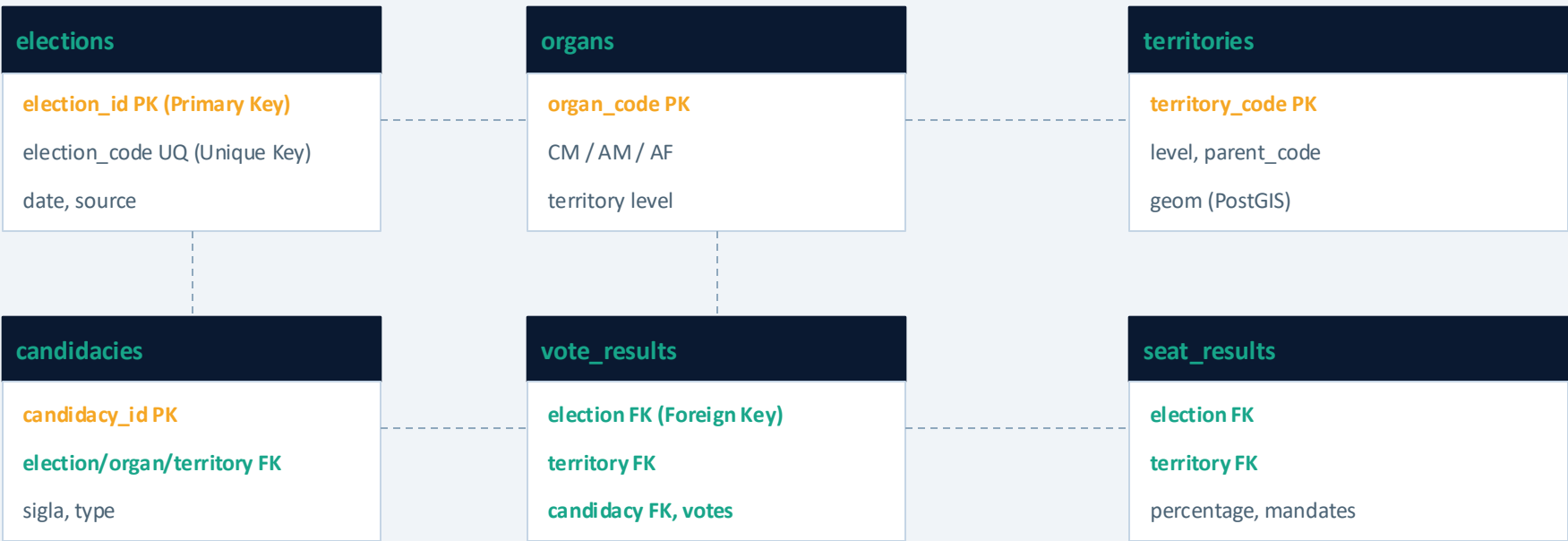
Explicit SQL via
psycopg2 - no ORM
hiding DB logic

Focus: database-centred · SQL explicit · Flask as thin presentation layer

System Architecture



Operational Data Model



Territory hierarchy: district → municipality → parish · Candidacy types: party, coalition, citizen group

ETL Pipeline

1

Parse CNE Excel

Dynamic column detection — 2013/17/21/25 differ in party blocks

2

Staging via COPY

Fast psycopg2 COPY into staging tables; rerunnable truncate

3

PL/pgSQL Transform

load_from_staging() resolves labels, codes, and constraints

4

Warehouse Refresh

refresh_from_operational() rebuilds dims + facts

5

CAOP GeoPackages

Reproject → EPSG:3763 · dissolve duplicates · update territories

Cleaning: territorial codes → 6 chars · numeric normalization · label resolution via mapa_anexo.xlsx

Data Loaded

Election	Vote rows	Elected
AL 2013	12,152	35,579
AL 2017	12,069	35,709
AL 2021	12,277	35,491
AL 2025	12,839	36,521

Total: **49,337**

Data Warehouse · Star Schema



Materialized views: mv_result_summary · mv_national_totals · mv_territory_winners → fast frontend queries

Functions, Procedures & Triggers

f(x) SQL Functions

vote_share()

turnout_rate()

dhondt_allocate()

⚙️ PL/pgSQL Procedures

load_from_staging()

refresh_from_operational()

⚡ Triggers

No negative turnout / votes / mandates
Blocks negative values for votes, mandates, and turnout.

Blank + Null $\leq$ Voters
No negative numbers allowed for any counts.

Vote & seat rows must match candidacy scope
Prevents insertion of negative data points.

D'Hondt in SQL

generate_series(1, seats) → rank with window function → pick top-n quotients

Analytical SQL

D'Hondt

Allocation

Mandate allocation as pure SQL using
`generate_series` + `RANK()`

Window Functions ×3

Analysis

Municipality rankings · turnout within district ·
seat-change trends

GROUP BY ROLLUP

Aggregation

Candidate subtotals by district → district totals
→ global total in one query

GROUP BY CUBE

Aggregation

Votes & mandates by candidate type × district
with all subtotals

FILTER & jsonb_agg

Advanced

Conditional aggregation + JSON output for
frontend endpoints

Historical SQL

Multi-year

Turnout changes · winner flips · vote-share
trends across 2013–2025

Spatial Layer & Flask Frontend

PostGIS Coverage (EPSG:3763)

Districts / Regions

20 / 20



Municipalities

308 / 308



Parishes

3241 / 3394



*ST_AsGeoJSON · ST_Transform · ST_SimplifyPreserveTopology → Leaflet GeoJSON
(308 features per election)*

Frontend Pages

/

Leaflet map · election selector · municipality colours by winner

/municipality/"MUNICIPALITY_NAME"

Turnout · results table · vote-share chart · seat chart · elected members

/compare

Political-family trends · seat changes · winner flips 2013–2025

Validation & Testing

D'Hondt Validation

EX: Agueda Municipal Council 2021

PPD/PSD.MPT



PS



CDS-PP



✓ Matches official CNE mandates

Flask Endpoint Tests

/

HTTP 200

/municipality/Agueda

HTTP 200

/compare

HTTP 200

Checks: staging · operational · warehouse · materialized views · D'Hondt · Flask

All 4 election cycles loaded · 49,337 DW facts · 308/308 municipality geometries · D'Hondt verified

Demo

1

Repository & SQL/ETL files

Structure, schema DDL, analytical queries, historical comparison SQL

2

D'Hondt query (ex: Agueda)

Run validation — verify official mandates PPD/PSD.MPT:4 · PS:2 · CDS-PP:1

3

Historical comparison page /compare

Vote-share trends 2013–2025 · municipalities where winner changed

4

Flask map /

Interactive Leaflet map · click a municipality · see results update

5

Municipality results page

Turnout · vote-share chart · seat chart · elected members list

Thank You

Questions?